

MATB24H3

Exercise Questions

Student Name: Your Name

Student Number: Your Student Number

2.1.1. Consider the set $X = \mathbb{Q} \setminus \{0\}$ and the map $\star$ is depicted by $a \star b = \frac{a}{b}$. Is $\star$ a binary operation?

2.1.2. Is $(\mathcal{P}(\mathbb{N}), \cap)$ a monoid? what is the identity element? How about $(\mathcal{P}(\mathbb{N}), \cup)$ for any arbitrary set X ?

2.2.1. Suppose d is square-free, so there is no integer m such that $m^2 = d$. Show that $(\mathbb{Q}(\sqrt{d}), +, \cdot)$ forms a field. What happens if $d = m^2$ for some m ?

2.2.2. Now, can we extend a finite field in the same way we extended an infinite field? If so, give an example.

2.2.3. Prove that $0_{\mathbb{F}}$, $1_{\mathbb{F}}$ and the additive inverse element $-a$ of any element $a \in F$ are unique.

2.2.4. Prove $(-a)(-b) = ab$ for all $a, b \in \mathbb{F}$.

2.2.5. Prove that the triple $(\mathbb{Z}_p, + \bmod p, \cdot \bmod p)$ is a field $\iff p$ is a prime. Use the fact that if two integers m, n are relatively prime, then there exist integers r and s such that $mr + ns = 1$.

2.2.6. Prove that the 2-tuple $(S, \circ)$, where S is the set of symmetries of a square and $\circ$ is the composition operation, does indeed form a group. Moreover, find the symmetry group of a rectangle.

2.2.7. Characterise all idempotent elements of $(\mathcal{M}_{2 \times 2}(\mathbb{R}), +, \cdot)$.

2.2.8. Prove that $\mathcal{C}(\mathcal{R})$ is a subring of any ring $(\mathcal{R}, +, \cdot)$.

2.2.9.